

Dual-core machine learning accelerator for attention mechanism

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I. INTRODUCTION

This report summarizes RTL design, synthesis, place-and-route, and verification for a dual-core attention accelerator. The following sections follow the course milestone flow (single core through sparsity-aware attention), merging the LaTeX report structure with the project outline document. Placeholder figures use `temp.png` where a dedicated asset is not yet checked in.

II. STEP 1: SINGLE CORE RTL, SYNTHESIS & PNR

Post-PnR layout, GLS waveform, and terminal output for the single-core design are shown below, with summary metrics in Table I.

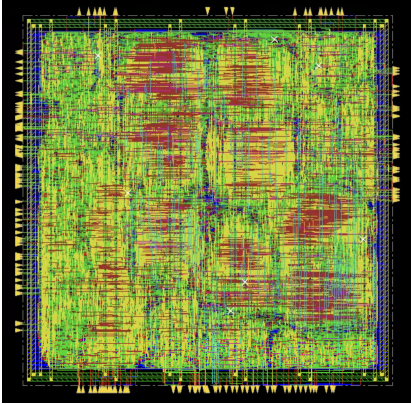


Fig. 1: Single-Core Layout (post-PnR).

TABLE I: Step 1 Results

	Power (VCD)	WNS	DRC
Single Core	50.74 mW	-0.014	0

III. STEP 2: OUTPUT NORMALIZATION

Behavioral simulation waveform and testbench overview (replace `temp.png` when the final waveform is finalized).

IV. STEP 3: HIERARCHICAL SYNTHESIS OF CORE

Memory macros and core integration: SRAM blocks (placeholders) followed by hierarchical core layout and GLS results.

```

xcelium> run
##### 0 data txt reading #####
##### K data txt reading #####
##### Open writing #####
##### Kmem writing #####
##### K loading to MAC #####
##### Execute #####
OK phase verification
(row) RTL      col0 col1 col2 col3 col4 col5 col6 col7      -63 -86 -96
[0] RTL      93    -20    -7    -7    94    -48    -45    -45    -63 -86 -96
    golden:   93    -20    -7    -7    94    -48    -45    -45    -63 -86 -96
    [OK]
[1] RTL      52    -21    -73    24    -5    -33    -41    -23
    golden:   52    -21    -73    24    -5    -33    -41    -23
    [OK]
[2] RTL      38    27    50    63    -28    21    -3    -55
    golden:   38    27    50    63    -28    21    -3    -55
    [OK]
[3] RTL      33    -74    17    125    -37    -17    -27    -51
    golden:   33    -74    17    125    -37    -17    -27    -51
    [OK]
[4] RTL      80    -10    -10    13    -138    -37    -90    -56
    golden:   80    -10    -10    13    -138    -37    -90    -56
    [OK]
[5] RTL      42    1    26    68    7    7    -8    -45
    golden:   42    1    26    68    7    7    -8    -45
    [OK]
[6] RTL      -19    48    99    -7    -1    23    46    -20
    golden:   -19    48    99    -7    -1    23    46    -20
    [OK]
[7] RTL      -63    41    128    -46    -1    78    74    37
    golden:   -63    41    128    -46    -1    78    74    37
    [OK]

PASS: all 8 x 8 results correct
Simulation complete via $finish(1) at time 111 NS + 0
/netlist/step1_th v 238      #10 $finish

```

Fig. 3: Single-Core GLS Terminal Output.



Fig. 4: Behavioral Simulation Waveform / TB.



Fig. 5: SRAM_64b (Qmem & Kmem) Layout (placeholder).

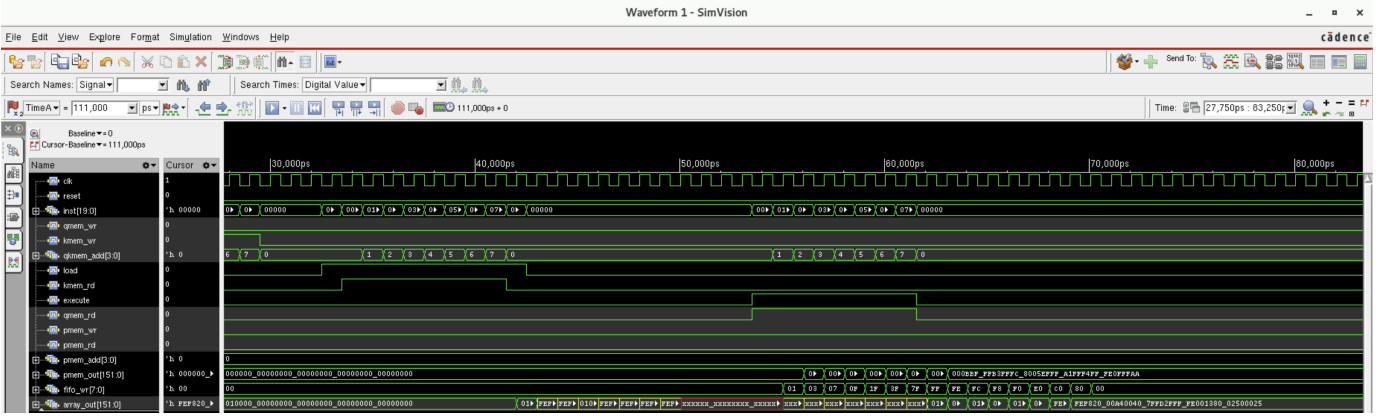


Fig. 2: Single-Core GLS Waveform.

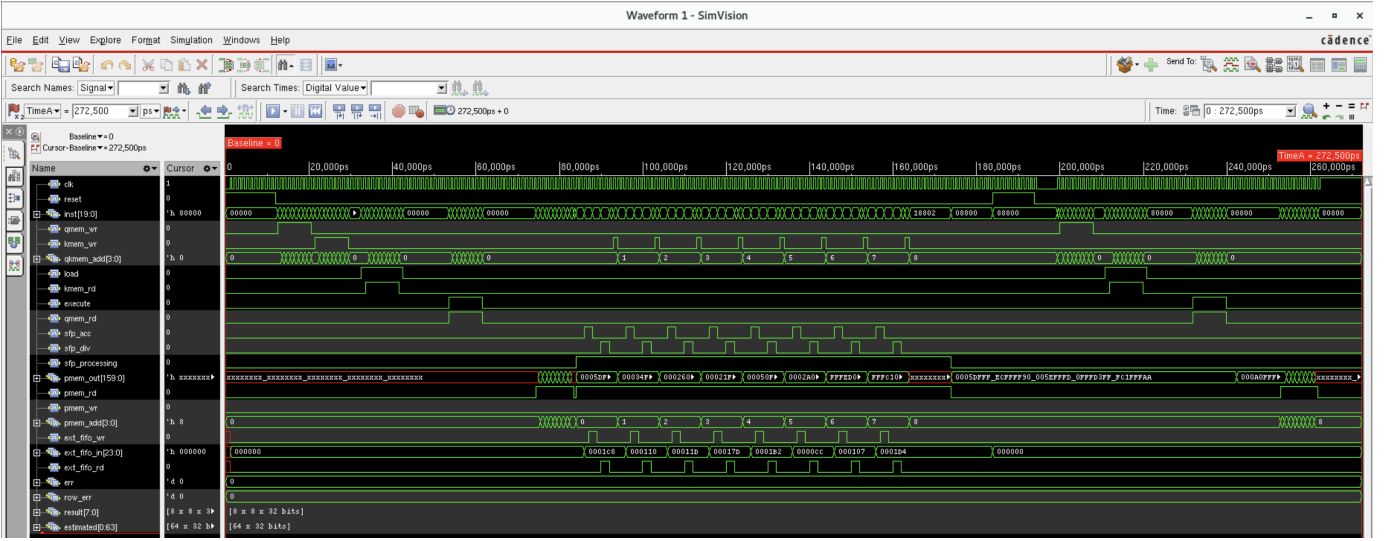


Fig. 8: Hierarchical Core GLS Waveform.



Fig. 6: SRAM_160b (Psum_mem) Layout (placeholder).

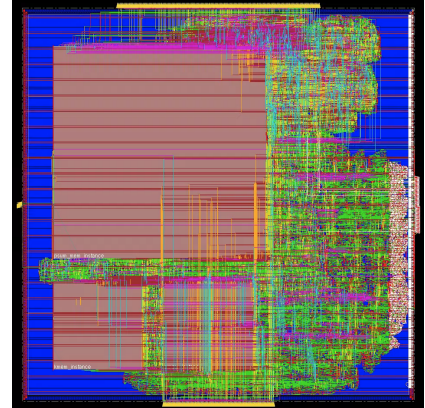


Fig. 7: Hierarchical Core Layout.

TABLE II: Step 3 Results

	Power (VCD)	WNS	DRC
Hier. Core	61.22 mW	—	—

A. Submodule and SRAM Layouts

B. Core Layout and GLS

V. STEP 4: HIERARCHICAL SYNTHESIS OF DUAL-CORE

This milestone includes both fullchip-style views (non-lockstep GLS) and hierarchical dual-core post-PnR results

```

#### execute ####
VN phase verification start (checking pmem content)
#### sample pmem content & compare to golden ####
(row) RTL: col0 col1 col2 col3 col4 col5 col6 col7
golden:
[0] RTL: 160 -103 155 232 -311 283 -242 -79
golden: 160 -103 155 232 -311 283 -242 -79
[OK]
[1] RTL: 238 85 98 127 334 179 54 75
golden: 238 85 98 127 334 179 54 75
[OK]
[2] RTL: -214 -333 -334 46 -430 -300 -281 -253
golden: -214 -333 -334 46 -430 -300 -281 -253
[OK]
[3] RTL: -517 -433 -354 -314 -526 -504 -309 -381
golden: -517 -433 -354 -314 -526 -504 -309 -381
[OK]
[4] RTL: -311 40 -233 -253 -196 -396 298 68
golden: -311 40 -233 -253 -196 -396 298 68
[OK]
[5] RTL: -419 -414 -467 -366 -456 -465 -371 -432
golden: -419 -414 -467 -366 -456 -465 -371 -432
[OK]
[6] RTL: -154 -21 -47 -73 140 -267 187 3
golden: -154 -21 -47 -73 140 -267 187 3
[OK]
[7] RTL: -201 24 -35 -201 -351 42 -58 -38
golden: -201 24 -35 -201 -351 42 -58 -38
[OK]
-----
PASS 8 rows x 8 cols all match estimated result
-----
Simulation complete via $finish(l) at time 275250 PS + 0
/netlist/core_tb.v:603 #10 $finish;

```

Fig. 9: Hierarchical Core GLS Terminal Output.

from the LaTeX report.

A. Fullchip / Non-Lockstep GLS (placeholders)



Fig. 10: Fullchip Layout (placeholder; replace with fullchip snapshot).



Fig. 12: Fullchip GLS Terminal Output, Non-Lockstep (placeholder).

B. Dual-Core Post-PnR (lockstep-style assets)

VI. STEP 5: OPTIMIZATION

Optimized dual-core figures (placeholders until PnR/GLS images are exported).

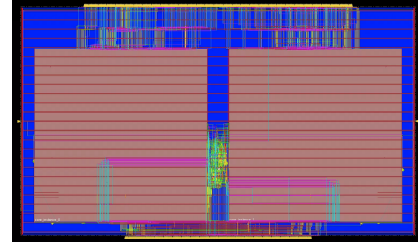


Fig. 13: Dual-Core Layout.

```

===== OVERALL SUMMARY =====
Mode: NON-LOCKSTEP, split 40-bit inst bus
clk0=1.000000Hz (core0) clk1=1.333000Hz (core1)
tick0 then tick1 per step - 1 posedge per core per operation
OK phase -- CORE0: 0 CORE1: 0 mismatch(es)
VN phase -- CORE0: 0 CORE1: 0 mismatch(es)
>>> ALL TESTS PASSED <<<
=====
Simulation complete via $finish(l) at time 575250 PS + 0
/netlist/fullchip_tb.v:714 #10 $finish;\r
xcelium>

```

Fig. 15: Dual-Core GLS Terminal Output.

TABLE III: Step 4 Results

	Power (VCD)	WNS	DRC
Dual Core	19.19 mW	—	—



Fig. 16: Optimized Dual-Core Layout (placeholder).



Fig. 18: Optimized Dual-Core GLS Terminal Output (placeholder).

TABLE IV: Step 5 Results

	Power (VCD)	WNS	DRC
Optimized	—	—	—

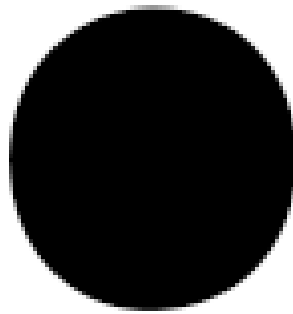


Fig. 11: Fullchip GLS Waveform, Non-Lockstep (placeholder).

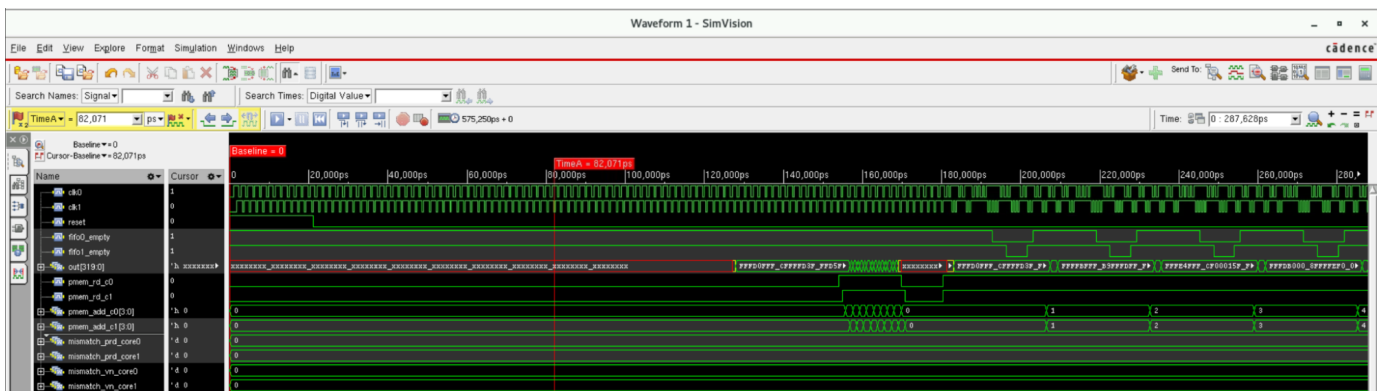


Fig. 14: Dual-Core GLS Waveform.

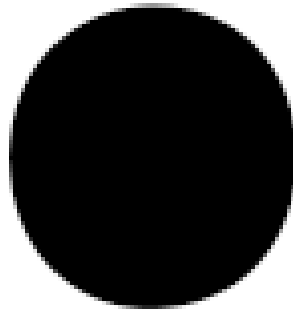


Fig. 17: Optimized Dual-Core GLS Waveform (placeholder).

VII. STEP 6: HIERARCHICAL SPARSITY-AWARE ATTENTION

Sparsity-aware dual-core figures (Word title: Sparsity-Aware Attention; placeholders until final images are added).

TABLE V: Step 6 Results

	Power (VCD)	WNS	DRC
Sparsity-aware	—	—	—



VIII. CONCLUSION

REFERENCES

- [1] E. Course, “Ece260b final project (placeholder),” Course project report, 2026, replace with real references; use cite in main.tex to list them.

Fig. 19: Sparsity-Aware Attention Dual-Core Layout (placeholder).

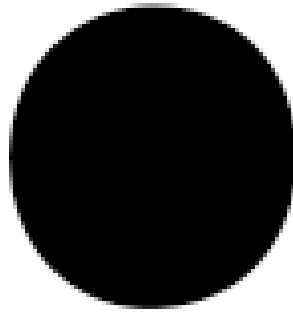


Fig. 20: Sparsity-Aware Attention Dual-Core GLS Waveform (placeholder).

APPENDIX

DESIGN NOTES (FROM PROJECT OUTLINE)

Divider architecture: pipelined long division vs. MCP “*l*” operator; synthesis favored long division; both RTL paths may remain with macro selection (`SFP_LONGDIV`, etc.).

Development flow: Makefile-driven build, version control (e.g., git), and documented directory layout.

Interface / controller: per-core control; example flow—matrix multiply, normalize, read out—with streamlined chip interface hiding unnecessary details from upper-level users.



Fig. 21: Sparsity-Aware Attention Dual-Core GLS Terminal Output (placeholder).